

AayuGranth (IP-SAKTI)

Comprehensive System Architecture, Technical Stacks & Feature Intelligence Flows

1. Executive Summary & Core Mission

AayuGranth (powered by the IP-SAKTI intelligence framework) is an institutional-grade, evidence-grounded AI assistant and legal compliance platform engineered specifically for the **Ayurveda & Traditional Healthcare Sector**. It bridges classical Vedic compendia (Charaka Samhita, Sushruta Samhita, Astanga Hridayam, Ayurvedic Pharmacopoeia of India) with contemporary statutory IP laws (Indian Patents Act 1970, Biological Diversity Act 2002/2023, Drugs & Cosmetics Act 1940/1945, and international frameworks including US FDA 21 CFR 111 and EU THMPD).

Unlike generic LLMs prone to legal hallucination, AayuGranth uses a **deterministic RAG pipeline, programmatic citation verifiers, and calibrated confidence scoring** over 6,343 vector-indexed statutory and classical chunks.

2. Full Technology Stack

Layer	Technology / Framework	Role & Technical Justification
Frontend	React 19, Vite, Tailwind CSS, Lucide Icons, Framer Motion	High-performance client UI with warm ivory / forest green legal aesthetic, responsive chat cards, active dossier banners, and zero latency state management.
Backend API	FastAPI (Python 3.14/3.11), Uvicorn (ASGI), Pydantic v2	Asynchronous high-concurrency REST endpoints with automated OpenAPI schemas, strict input validation, and robust stage-wise exception telemetry.
Database	MongoDB Atlas (Motor Async Client), Database: <i>ipsakti</i>	Stores 6,343 legal chunks with 1024-dim dense vectors, product passport documents, active dossiers, and human escalation queues.
Embeddings	BAAI/bge-m3 (1024 Dimensions, L2-Normalized)	State-of-the-art multilingual legal & technical dense embedding model running locally in PyTorch/SentenceTransformers with prewarmed lifecycle.
Vector Search	Atlas \$vectorSearch + In-Memory Cosine Fallback	True mathematical cosine similarity calculation without artificial or fabricated score heuristics.
LLM Engine	Google Gemini 3.6 Flash / LangChain / LangGraph	Grounded generation with strict system prompts, JSON structured schemas, and fallback deterministic statutory synthesis.
Verification	Citation Verifier & Calibrated Confidence Engine	Pure-Python assertion-to-chunk ID matching, score aggregation, and automatic abstention routing on missing evidence.

3. Core Features & In-Depth Technical Workflows

The platform is structured into 8 production-grade functional modules:

Feature 1: Institutional Legal & IP Chatbot (AskAayuGranth)

Description: Provides interactive, legally cited Q&A; on Ayurvedic patentability, traditional knowledge citations, ABS biodiversity obligations, and manufacturing licenses with zero hallucination.

Technical Flow & Architecture:

1. **User Input:** User types query in `AskAayuGranth.jsx` (optionally passing Product Passport context).
2. **API Route:** Client sends `POST /api/v1/ask/` with query and jurisdiction.
3. **Domain Scope Filter:** `services/out_of_scope_filter.py` screens query. Unrelated topics (weather, general trivia) immediately receive a domain advisory response.
4. **Embedding & Retrieval:** `services/similarity_engine.py` embeds query using BAAI/bge-m3, queries MongoDB Atlas, computes true cosine similarities, and deduplicates top statutory chunks.
5. **Grounded Generation:** `services/rag_pipeline.py` formats prompt with retrieved statutory context and invokes Gemini (or executes deterministic statutory synthesis on rate limit).
6. **Verification & Scoring:** `citation_verifier.py` verifies claimed chunk IDs; `confidence_engine.py` computes confidence score (High/Moderate/Low/Abstain).
7. **Client Render:** Returns structured JSON. Frontend renders a text-first Research Response Card with summary, key findings, and expandable verified citations.

Feature 2: IP-SAKTI Digital Product Passport (Credential & Dossier)

Description: Serves as a digital credential and evidence-backed regulatory dossier for Ayurvedic formulations. Includes a virtual security-banded ID card, QR verification, 6-domain audit, explainability modal, and dual download workflows.

Technical Flow & Architecture:

1. **Data Generation:** `POST /api/v1/passport/generate` evaluates ingredients against classical formularies, Section 3(p) exclusions, and ABS schedules.
2. **Virtual Passport Card:** Displays Qualitative Status (VALIDATED / NEEDS REVIEW), Corridor badges, and dynamic verification QR code.
3. **4-Step Explainability Chain:** Traces Product Signal → Retrieved Statutory Evidence → Legal Interpretation → Computed Posture.
4. **Dual Downloads:** 'Passport Only' prints cleanly isolated card; 'Full Data' exports 6-domain PDF dossier or JSON dataset.

Feature 3: Section 3(p) Patentability & Prior-Art Novelty Screener

Description: Evaluates Ayurvedic formulations against Section 3(p) of the Indian Patents Act, 1970 (which bars patenting traditional knowledge or simple aggregations of known properties).

Technical Flow & Architecture:

1. **NER Extraction:** `services/ner_service.py` extracts botanical names, Sanskrit synonyms, and extraction methods.
2. **TKDL Overlap Analysis:** `routes/tk_engine.py` checks classical treatises (Charaka, Sushruta, AFI) to identify prior-art disclosures.
3. **Synergy Assessment:** If formulation demonstrates novel extraction kinetics or unexpected therapeutic synergy, flags patentable pathway subject to NBA approval.

Feature 4: Access & Benefit Sharing (ABS) Biodiversity Compliance Engine

Description: Analyzes obligations under Biological Diversity Act 2002, 2023 Amendment Act, and the 2025 ABS Regulations.

Technical Flow & Architecture:

1. **Entity Screening:** Checks applicant jurisdiction (Indian entity vs Section 3(2) foreign-controlled entity).
2. **Turnover Fee Calculator:** Applies 2025 ABS brackets (Up to ₹5 Cr: Nil | ₹5-50 Cr: 0.2% | ₹50-250 Cr: 0.4% | >₹250 Cr: 0.5%).
3. **Form Routing:** Maps compliance requirements to Form-1 (Commercial Access), Form-2 (Transfer), or Form-3 (IP/Patent Approval).

Feature 5: AYUSH Regulatory Pathway & Licensing Navigator

Description: Determines exact licensing classification under Drugs & Cosmetics Rules 1945 (Rule 158-B) and Schedule T GMP requirements.

Technical Flow & Architecture:

1. **Classification Engine:** `routes/regulatory.py` parses ingredient ratios to categorize into Classical Ayurvedic Medicine, Ayurvedic Proprietary Medicine, or Plant Extract Formulation.
2. **Clinical Exemption Mapping:** References 54 authoritative books in the First Schedule to exempt classical formulations from fresh Phase I-III human clinical trials.
3. **Schedule T Checklist:** Verifies heavy metal limits (Lead, Arsenic, Mercury, Cadmium) and microbial limits.

Feature 6: Patent Document Readiness & Drafting Auditor

Description: Audits uploaded patent drafts and formulation specifications for Indian Patent Office (IPO) and PCT filing readiness.

Technical Flow & Architecture:

1. **Document Parsing:** `services/document_parser.py` extracts text from PDF, DOCX, DOC, and TXT files using PyPDF and docx2txt.
2. **Section 10(4) Checklist:** Audits Title, Abstract, Background, Complete Specification, Claims, and Biological Material Source Disclosure.
3. **Readiness Score:** Computes a 0–100 Readiness Score with itemized recommendations on missing legal elements.

Feature 7: International Export Corridor & Cross-Border Compliance

Description: Analyzes destination market requirements for cross-border Ayurvedic export (US, EU, UK, GCC).

Technical Flow & Architecture:

1. **US FDA Pathway:** Navigates Dietary Supplement Health and Education Act (DSHEA) and 21 CFR 111 cGMP compliance.
2. **EU THMPD Pathway:** Verifies 30-year traditional use evidence (including 15 years in EU) for Traditional Herbal Medicinal Product registration.
3. **Prohibited Substance Screener:** Screens herbs against US FDA Import Alert red-lists (e.g., Aristolochia, Ephedra).

Feature 8: Escalation & Human-in-the-Loop IP Facilitator Workflow

Description: Seamlessly bridges AI intelligence with human IP attorneys and Ayurvedic regulatory experts when automated confidence is low or complex novelty questions arise.

Technical Flow & Architecture:

1. **Abstention Trigger:** When `compute_confidence` flags `should_abstain: True`, the assistant avoids speculation.
2. **Docket Creation:** `POST /api/v1/escalate/` creates an audit ticket in MongoDB with query, product ID, and timestamp.
3. **Facilitator Assignment:** Generates unique tracking ID (ESC-XXXXXX) for human attorney review.

4. Benchmark & Verification Test Results

Query / Test Scenario	Status	Confidence	Statutory Grounding & Result
What is Section 3(p) of the Indian Patents Act?	200 OK	Moderate (0.69)	Cites Form-3 NBA Rules and clarifies exclusions for traditional knowledge aggregations.
Can a traditionally known Ayurvedic formulation be patented?	200 OK	Moderate (0.67)	Explains non-patentability of known recipes; highlights novelty & synergistic extraction criteria.
What are the ABS obligations for an Ayurvedic formulation?	200 OK	Moderate (0.67)	Cites 2025 ABS Regulations fee schedule (0.2% - 0.4% above ₹5 Cr turnover).
What are the licensing requirements for an Ayurvedic medicine?	200 OK	Moderate (0.69)	Cites Section 7 of Biological Diversity Act and D&C; Schedule T manufacturing standards.
What is the weather today? (Out of Scope)	200 OK	High (0.95)	Out-of-scope domain filter triggers guidance message on Ayurveda IP scope.

5. Summary & Architectural Rigor

AayuGranth represents a paradigm shift in specialized legal-tech AI. By uniting dense vector retrieval over authenticated statutory texts with strict citation verification and calibrated confidence engines, the platform guarantees that every legal assertion, patent claim, and regulatory obligation presented to researchers and manufacturers is verified, reliable, and actionable.